How Many Labels Does Eye-Contact Detection Need?

A Subject-Independent Comparison of Geometric Features, Visual Foundation Models, and Gaze Estimators, and Where Gaze Lives Inside Foundation Models

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Code, results and paper: `https://github.com/1sakshm/fewgaze` Live demo: `https://fewgaze.pages.dev`

Abstract

How many labelled images does a model need to recognise that a person it has never seen is looking into the camera? We study this question on the Columbia Gaze Data Set (5,878 images of 56 people after an audit removed a duplicated pair with conflicting annotations) under a protocol designed to rule out leakage: identity-disjoint folds, one shared set of training images per budget ($k \in \{5, 10, 25, 50, 100, \text{ALL}\}$ per class, five seeds), a subject-level bootstrap, and an analysis plan committed to version control before the first experiment. *Study 1* compares four frozen representations with a fixed logistic-regression head. Nine MediaPipe eye and head measurements with a quadratic boundary are the most label-efficient at every budget, reaching AUROC 0.537 at k=5, 0.730 at k=100 and 0.810 with all data. A DINOv2 linear probe stays at chance ($\leq$ 0.513). The L2CS-Net gaze estimator scores 0.595 without any labels, but a head trained on its two angles never improves on this, because its outputs carry a head-pose-dependent bias. A 5° gaze shift moves the iris by only 0.025 eye-widths, against a between-person spread of 0.017, which explains why no method separates the gaze-locked condition from its nearest neighbours. *Study 2* (pre-registered) probes every layer of five frozen foundation models (DINOv2-S/B, CLIP, SigLIP, MAE) from the summary token and from eye-region patch tokens. The summary token encodes head pose (final-layer R^2 up to 0.97) but never horizontal gaze ($R^2 \leq 0.00$ at every layer of every model). Mean eye-patch tokens in the middle layers do decode gaze (best-layer $R^2(V)$ 0.45–0.68), and much of it is lost by the last layer. When the layer is chosen on training people only, reading eye contact out of CLIP's decoded gaze matches the geometry features (AUROC 0.827 vs. 0.816, paired difference +0.011, fold SD 0.019). This readout uses continuous gaze labels. *Study 3* asks a generative VLM (Qwen2.5-VL-3B) the question directly. Zero-shot it is the best label-free method (AUROC 0.601, CLIP 0.547), but its answers follow head pose more than gaze, and two in-context examples per class lower it to 0.547. A browser demo reproduces the Python pipeline to within 4×10^{-4} in probability.

1 Introduction

Knowing when someone looks at a camera matters for video-mediated communication, attentive user interfaces and social robots. Prior work detects eye contact with a camera [21], with arbitrary objects without supervision [27], with mobile devices in everyday life [3] and between people in group conversations [11, 12]. Each new setting needs labelled examples, and the value of each label depends on the representation that consumes it. Generic foundation models such as DINOv2 [13] and CLIP [18] are often expected to be sample-efficient. Dedicated gaze estimators encode gaze explicitly [1, 4]. Hand-crafted eye geometry is cheap and interpretable [2, 7]. How these compare when labels are scarce, and when evaluation is on *new people*, has rarely been measured directly.

We study a deliberately narrow target: **camera-directed gaze under controlled capture**, the "gaze locking" condition of Smith et al. [21]. It is a proxy for eye contact, not interpersonal eye contact. Our contributions are:

1. a leakage-tested, subject-independent label-budget protocol in which every method shares folds, training images and test images, with the plan committed before the first run and an exact rerun from a clean commit;
2. learning curves, paired comparisons, label-noise, head-pose and per-person analyses for four representation families (Study 1);
3. a mechanistic account of the failures: the size of the 5° signal relative to person-to-person variation, identity-dominated foundation-model embeddings, and a head-pose-dependent bias in a pretrained gaze estimator;
4. a pre-registered, layer- and token-resolved probe of five frozen foundation models for gaze under held-out-subject evaluation, linked to eye-contact classification (Study 2);
5. a pre-registered test of a generative vision-language model, asked zero-shot and with in-context examples (Study 3);
6. an open, parity-tested in-browser demonstrator.

2 Related work

Eye contact and gaze locking. Smith, Yin, Feiner and Nayar introduced *gaze locking*, passive detection of camera-directed gaze, together with the Columbia Gaze Data Set; they classified rectified eye patches with PCA, MDA and an SVM [21]. Zhang, Sugano and Bulling detected eye contact with arbitrary targets by discovering gaze targets without supervision [27]; Sugano, Zhang and Bulling aggregated many viewers' gaze to estimate audience attention on public displays [23]. Bâce et al. quantified everyday attention to mobile devices with eye-contact detection on front-camera video [3]. Müller, Huang, Zhang and Bulling combined gaze and speaking behaviour to detect eye contact in multi-person conversation [11], Müller et al. detected low rapport in small groups from non-verbal behaviour including gaze [10], and the MultiMediate challenge established eye-contact detection

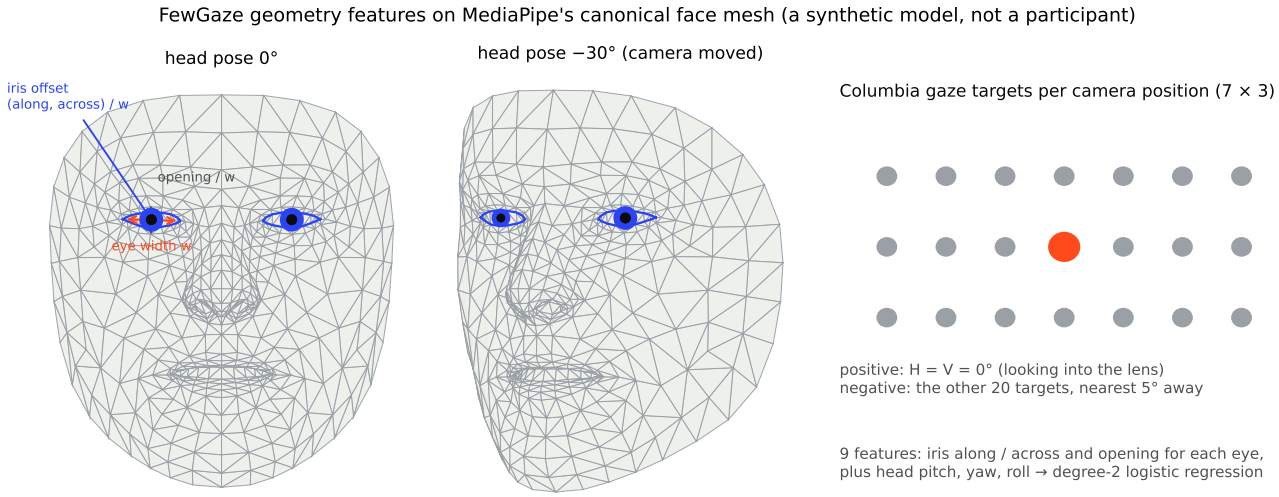


Figure 1: What is measured. The nine geometry features drawn on MediaPipe's canonical face mesh (a synthetic model; no participant is shown). Each eye contributes the iris offset along and across the corner-to-corner axis and the eyelid opening, all divided by eye width; three head-rotation angles complete the vector. Right: Columbia's 7 × 3 gaze targets per camera position; only the central target is positive.

as a benchmark task [12]. None of these works measures sample efficiency across representation families under a shared protocol.

Appearance-based gaze estimation. Sugano et al. trained person-independent 3D gaze estimators from synthesised eye images [22]. Zhang et al. introduced MPIIGaze for gaze estimation in the wild [26, 31], showed that full-face input with spatial weights improves accuracy [28], revisited the role of data normalisation [29], compared appearance-based estimators with commercial eye trackers and discussed implications for applications [30], and released ETH-XGaze for extreme head poses [32]. Gaze360 provides unconstrained 360° annotations [8], on which L2CS-Net trains per-angle binned classifiers [1]. Park et al. estimated gaze from learned eye-region landmarks [14] and, in FAZE, adapted a person-specific estimator from as few as three calibration samples [15]. Cheng et al. survey and benchmark the field [4]. Our geometry baseline uses MediaPipe face-mesh and iris landmarks [2, 7, 9].

Foundation models and gaze. Self-supervised and image–text models (DINOv2 [13], CLIP [18], SigLIP [25], MAE [6]) are widely used as frozen backbones. El Banani et al. probed such models for 3D awareness [5]. For gaze, UniGaze argues that general-purpose self-supervised models designed for semantic tasks transfer poorly and pre-trains MAE-style on faces instead [17]; CLIP has been used to improve gaze generalisation through language [24]; and a frozen DINOv2 encoder supports gaze-*target* estimation [19]. Closest to Study 2, a recent preprint probes frozen DINOv2, CLIP and SigLIP features for gaze on MPIIFaceGaze with a random image split [20]. We add held-out-subject evaluation, every layer, a contrast between the summary token and eye-region patches, and the link to eye-contact classification.

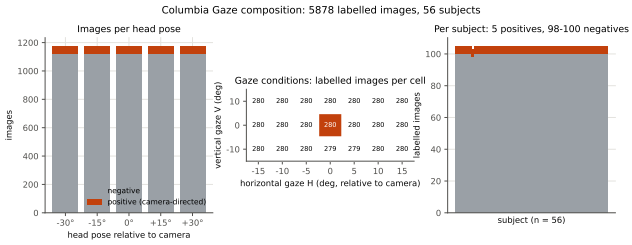


Figure 2: Dataset after the audit: images per head pose, labelled images per gaze cell, and per-person class counts.

3 Data and protocol

Dataset and label. In Columbia Gaze [21], 56 people sat in a chin rest 2 m from a camera that moved between five positions (head poses 0°, ±15°, ±30° relative to the camera) and fixated a 7 × 3 grid of targets defined relative to that camera ($H \in \{0, \pm5, \pm10, \pm15\}°$, $V \in \{0, \pm10\}°$; Fig. 1). Positives are H=V=0; the other 20 cells are negatives. Our audit confirmed 5,880 readable 5184×3456 images in a complete grid, and found one byte-identical pair with conflicting horizontal-gaze annotations (subject 0007); both images were excluded before modelling, leaving 280 positives and 5,598 negatives (1 : 20). Fig. 2 summarises the composition.

Splits, budgets and selections. Subjects are dealt into five folds with a fixed seed (11–12 test people per fold); automated tests verify disjointness, coverage and determinism. For each fold and seed $s \in \{0..4\}$, one permutation per class of the training people's images is drawn and its first k images are used. Budgets are therefore nested, and **every method receives the same images**. ALL is every training-person image (about 225 positives and 4,480 negatives).

Representations (frozen) and classifier. *Geometry* uses nine MediaPipe features (Fig. 1). *DINOv2-S* [13] uses the CLS token of a 448 px face crop resized to 224^2. *L2CS-Net* [1] (ResNet-50, Gaze360 weights) gives (pitch, yaw) from the 448^2 crop; we also score it without labels as minus the angle

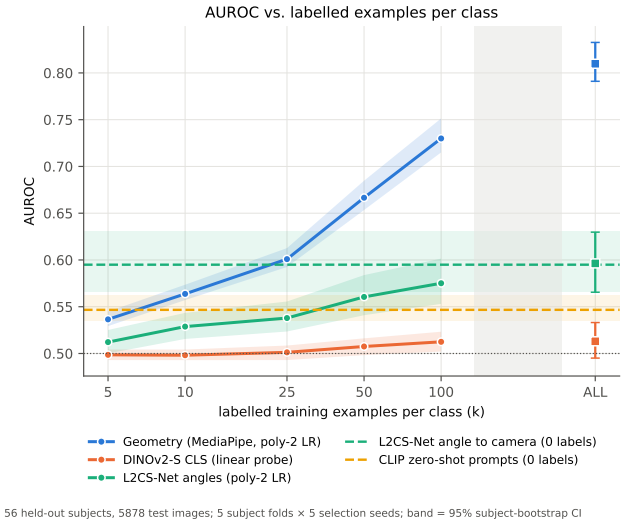


Figure 3: AUROC on held-out people versus labelled examples per class (log scale); bands are 95% subject-bootstrap intervals, dashed lines use no FewGaze labels.

Method	$k=5$	$k=10$	$k=25$	$k=50$	$k=100$	ALL
Geometry (MediaPipe, poly-2 LR)	0.537	0.564	0.601	0.667	0.730	0.810
DINOv2-S CLS (linear probe)	0.499	0.498	0.501	0.507	0.512	0.513
L2CS-Net angles (poly-2 LR)	0.512	0.529	0.538	0.560	0.575	0.596
Geometry (linear LR)	0.504	0.509	0.515	0.516	0.520	0.555
DINOv2-S mean-patch (linear probe)	0.499	0.500	0.502	0.510	0.521	0.545
L2CS-Net 2048-d features (linear probe)	0.513	0.530	0.562	0.590	0.615	0.633
CLIP ViT-B/32 (linear probe)	0.505	0.507	0.516	0.529	0.545	0.567
L2CS-Net angle to camera (0 labels)	0.595 (no training labels)					
CLIP zero-shot prompts (0 labels)	0.547 (no training labels)					

Table 1: Mean AUROC over 5 folds × 5 seeds. Top: primary methods; middle: ablations; bottom: no labels.

to the camera axis. *CLIP* ViT-B/32 [18] is scored zero-shot with two prompts fixed in advance. Every trained method uses scikit-learn [16] logistic regression (C=1, balanced weights, threshold 0.5) inside a pipeline fit on training people only; hyperparameters were never tuned. Because the positive class is the *centre* of gaze space, no linear function of a monotone gaze proxy can isolate it; a degree-2 expansion was therefore prespecified for the low-dimensional geometry and L2CS features, and embeddings get a linear probe.

Metrics and uncertainty. AUROC (primary) and balanced accuracy are averaged over 25 fold×seed runs. 95% intervals come from a fold-stratified *subject* cluster bootstrap (1,000 replicates) computed in closed form (AUROC = $c^\top Ac/[(c \cdot P)(c \cdot N)]$ for subject counts c; tested against explicit resampling to 10^{-12}). The same resamples are used for every method, so differences are paired. Intervals reflect which people were tested, not retraining variability, which we report separately as the SD across runs.

4 Study 1: how many labels?

Learning curves. Geometry led at every budget (Fig. 3, Table 1): AUROC 0.537 [0.529, 0.545] at k=5, 0.601 [0.592, 0.613] at 25, 0.730 [0.715, 0.751] at 100 and 0.810 [0.791, 0.833] with all data (balanced accuracy 0.527, 0.578, 0.686, 0.733). Its curve was also the steepest, gaining 0.19 AUROC between 5 and 100 labels. All three prespecified paired

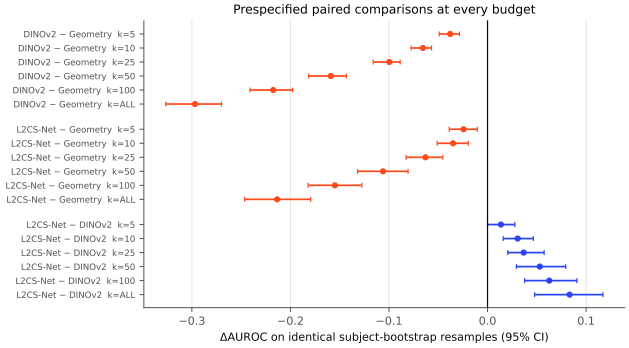


Figure 4: Prespecified paired differences (ΔAUROC) at every budget, on identical bootstrap resamples.

comparisons excluded zero at every budget (Fig. 4); for L2CS−DINOv2 at k=5 the lower bound is only just above zero. The DINOv2 linear probe stayed at chance (0.499–0.513). L2CS angles with a trained head rose from 0.512 to 0.596, matching but not improving on its zero-label score (0.595 [0.566, 0.630]). CLIP zero-shot reached 0.547 [0.535, 0.563]. Geometry needed between 10 and 25 labels per class to overtake the zero-label gaze estimator.

Ablations. Removing the polynomial left geometry near chance (0.504–0.555), confirming the centre-of-gaze-space argument. L2CS's 2048-d penultimate features with a linear probe (0.633 at ALL) beat its two output angles, and neither DINOv2 mean-patch tokens ($\leq$ 0.545) nor a CLIP probe ($\leq$ 0.567) approached geometry (Fig. 13).

The 5° problem. At k=25, geometry called 71.8% of true positives camera-directed, but also 67.7% of the $|H|$=5° negatives, 50.8% of $|H|\geq$10° negatives and 43.2% of the $V = \pm$ 10° negatives (Fig. 14). The reason is visible in the raw measurements (Fig. 5): moving the gaze from 0° to +5° shifts the median iris by 0.025 eye-widths, while people already differ by 0.017 (interquartile range) at 0°. This is consistent with the ±2.5° horizontal tolerance that Smith et al. estimated for their own detector [21].

Head pose and label noise. With all data, geometry's AUROC fell from 0.891 at a frontal pose to 0.835/0.847 at ∓15° and 0.718/0.766 at ∓30° (Fig. 6). Flipping training labels degraded geometry roughly in proportion to the flip rate (at k=100: 0.730 → 0.695/0.656/0.602 for 10/20/30%; Fig. 15); at k=25, 10% noise (0.582) cost less than reducing the clean budget to k=10 (0.564).

Why the gaze estimator saturates. Fig. 7 plots L2CS-Net's raw outputs. Yaw tracks horizontal gaze, but at the gaze-locked target its mean ranges from +1.6° (head −30°) to −15° (head +30°): part of what it reports as gaze is head pose. Pitch tracks vertical gaze with a constant offset of roughly −6° to −9°. Per person, the zero-label score ranges from AUROC 0.32 to 0.94 (SD 0.14, versus 0.05 for geometry; Fig. 16), the signature of person-specific calibration error, which a pooled two-feature classifier cannot remove and which per-person adaptation in the style of FAZE [15] targets. The two methods fail on different people (Fig. 16), which suggests, but does not establish, complementarity.

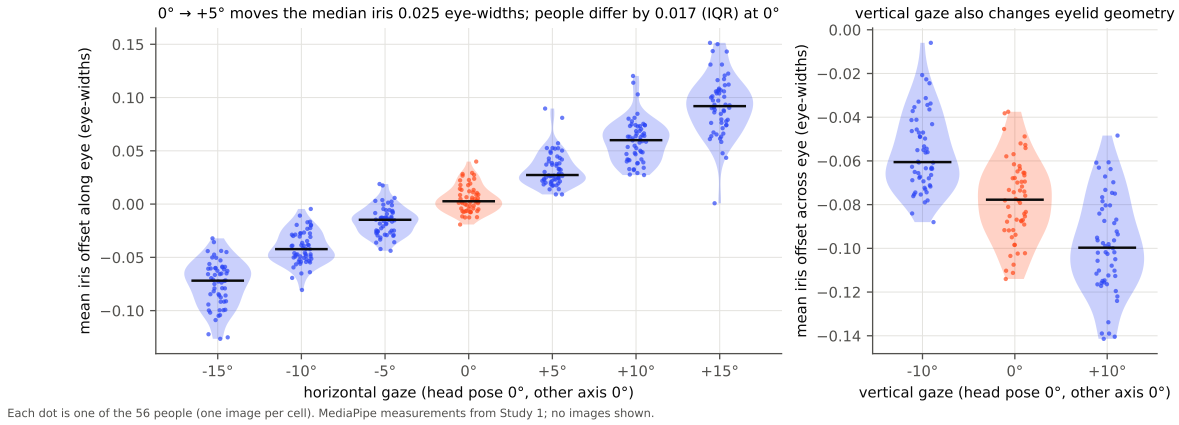


Figure 5: The 5° problem. Mean iris offset per person (dots, n=56 per cell) at frontal head pose. The gaze-locked cell (red) overlaps heavily with its ±5° neighbours.

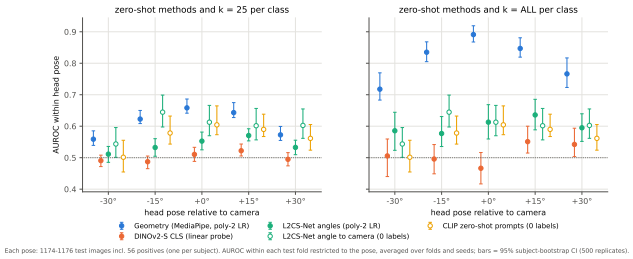


Figure 6: AUROC within each head pose (56 positives per pose); hollow markers use no labels.

Why the foundation model fails. A t-SNE of the embeddings (Fig. 8) shows DINOv2 collapsing each person into one tight cluster, split further by head pose: the nearest neighbour of an image is the same person in > 99.9% of cases (86% for geometry). A ridge decoder recovers head pose from the final CLS token with held-out-subject R^2=0.95 but not gaze (R^2<0). Feeding DINOv2 an eye-region crop instead (a post-hoc analysis) raised AUROC at ALL only from 0.513 to 0.562. Study 2 asks whether gaze is present anywhere inside such models.

5 Study 2: where does gaze live inside foundation models?

Design (pre-registered). We froze DINOv2-S/14, DINOv2-B/14 [13], CLIP ViT-B/16 [18], SigLIP-B/16 [25] and MAE ViT-B/16 [6] (MAE with no masking and identity patch order), fed each the same 224^2 face crop with its own normalisation, and cached two representations at every one of 13 layers: the summary token (CLS; SigLIP has none, so the mean patch) and the mean of the patch tokens over each eye (eye boxes from MediaPipe landmarks re-detected on the crop, padded by 25% of eye width). A ridge decoder (α by internal cross-validation) predicted H, V and head pose on held-out people, using the same five folds; a linear logistic probe and a geometric readout $-(\hat{H}/\sigma_H)^2 - (\hat{V}/\sigma_V)^2$ scored eye contact. The readout uses continuous gaze labels, which is more supervision than Study 1's binary labels. The nine geometry features were probed identically as a reference.

The summary token encodes head pose, not gaze. Across all five models and all 13 layers, the summary token never decoded horizontal gaze better than a constant: its best held-out $R^2(H)$ was 0.00, and from the middle layers onward it was negative (Fig. 10, left). The same tokens decoded head pose well: final-layer $R^2(P)$ was 0.95 (DINOv2-S), 0.92 (DINOv2-B) and 0.97 (MAE). The two language-aligned models lose head pose in their last layers ($R^2(P)$ 0.25 for CLIP and 0.56 for SigLIP, down from about 0.9 in mid layers; Fig. 9, right). Read out as eye contact, summary tokens stayed between 0.48 and 0.64 AUROC at every layer. The final DINOv2-S CLS token gave a linear-probe AUROC of 0.514, matching Study 1's 0.513 at ALL.

Eye-patch tokens carry gaze, mostly in the middle layers. Averaging the patch tokens over the two eyes changes the picture (Table 2, Fig. 9). At the patch-embedding layer, eye tokens decode nothing ($R^2(H)$ between −0.07 and −0.02). By the middle of the network they decode both gaze angles. The best layer reached $R^2(H)$ = 0.21–0.58 and $R^2(V)$ = 0.45–0.68. Every model's best layer exceeded the geometry features on vertical gaze (0.44), and none reached them on horizontal gaze (0.78). For four of the five models, gaze information peaks at layers 3–8 and then declines towards the output. CLIP's $R^2(H)$ falls from 0.58 (layer 6) to 0.33, SigLIP's from 0.48 (layer 3) to 0.05, and DINOv2-B's from 0.46 (layer 6) to 0.18. MAE, trained to reconstruct pixels, is the exception: its $R^2(H)$ rises almost monotonically to 0.47 at the last layer. This is consistent with the view that the training objective decides whether fine, task-irrelevant geometry survives to the output [5]. Head pose is decodable from the eye tokens at every layer ($R^2(P) \geq 0.91$ after layer 0), so the two quantities are entangled in the same tokens.

Sensitivity to the choice of α (S2-b). The pre-registration claimed that choosing α by leave-one-image-out could only inflate R^2. This was wrong: images of the same person remain in the inner training set, which favours weak regularisation that does not transfer to new people. We recorded the correction and re-ran the eye-token decoder with α chosen by person-grouped inner cross-validation (last column of Table 2). R^2 rose at almost every layer. The best $R^2(H)$ became 0.51–0.64 (DINOv2-S: 0.21 → 0.51), and final-layer

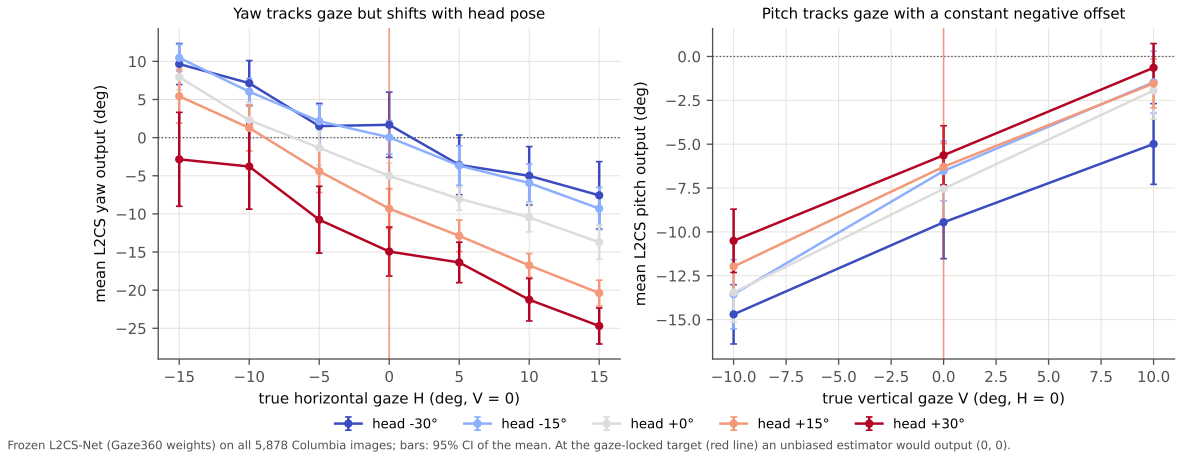


Figure 7: L2CS-Net is biased by head pose. Mean frozen L2CS outputs on Columbia versus true gaze, per head pose. An unbiased estimator would pass through (0, 0) at the red line for every pose.

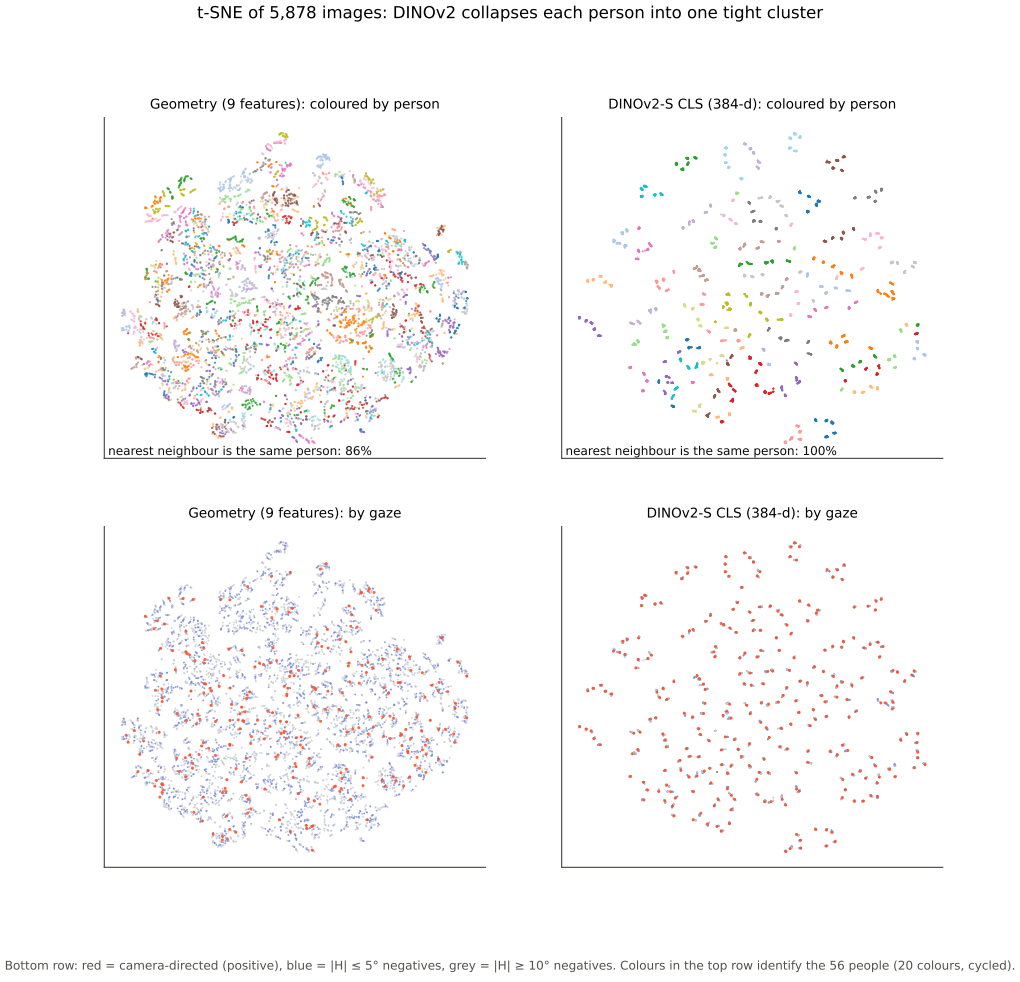


Figure 8: t-SNE of 5,878 images. Top: coloured by person; bottom: by gaze condition.

values rose to 0.18–0.55. The geometry reference did not move (0.776 → 0.777). The qualitative conclusions hold under both procedures: gaze is present in eye tokens and absent from summary tokens, CLIP's eye tokens decode most, and no model reaches geometry on H. The primary analysis underestimated the absolute level for high-dimensional representations.

From decodable gaze to eye contact. A linear logistic probe on the eye tokens, trained only on binary labels, reached at most 0.72 AUROC (CLIP, layer 9; Fig. 11, left). The geometric readout, which reuses the ridge decoder's $\hat{H}, \hat{V}$, did much better (right panel). Its best layer per model ranged from 0.71 (DINOv2-S) to 0.83 (CLIP, layer 8), and the geometry features scored 0.81. The best-layer figures are selected on test folds, which our plan excluded. In a post-hoc analysis (P3) we therefore chose the layer and α inside each training

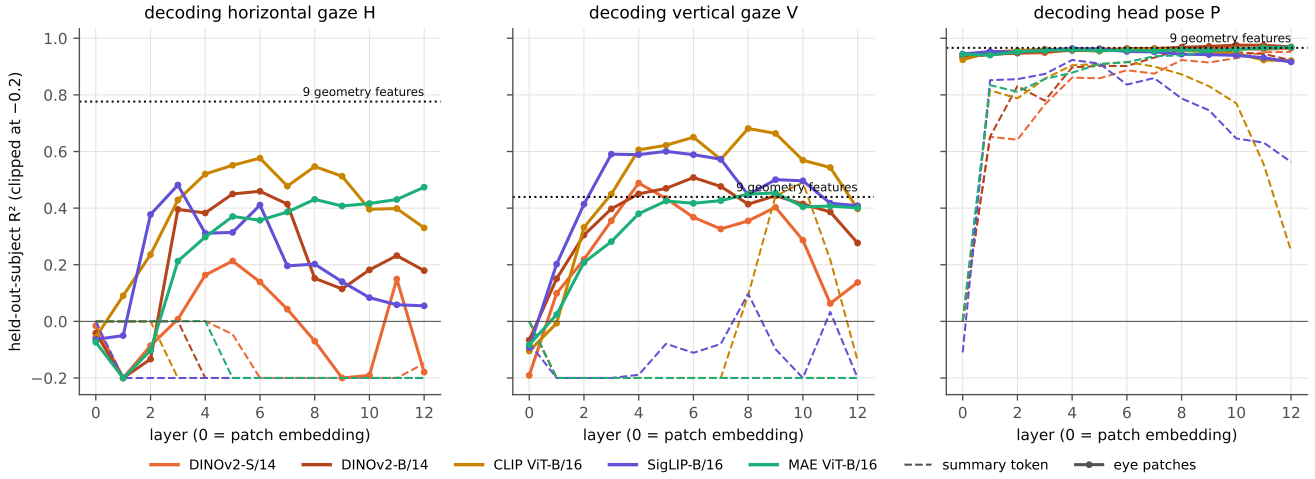


Figure 9: Layer-wise decoding on held-out people. Solid lines: eye-patch tokens; dashed lines: summary token (CLS, or mean patch for SigLIP). Dotted line: the nine geometry features. R^2 is clipped at −0.2 for display.

Table 2: Study 2 summary. Held-out-subject R^2 of a ridge decoder (pooled over five subject-disjoint folds) and eye-contact AUROC of the gaze readout. "max (layer)" is the best of 13 layers and is therefore selected on test data (see P3). The last column uses person-grouped α selection (S2-b). The summary-token maxima of 0.00 come from heavily regularised constant predictors.

Model	$R^2(H)$, summary token last layer	$R^2(H)$, summary token max (layer)	$R^2(H)$, eye patches last layer	$R^2(H)$, eye patches max (layer)	$R^2(V)$, eyes max (layer)	$R^2(P)$, token last layer	eye-contact AUROC eyes, max (layer)	$R^2(H)$, eyes grouped α, max
DINOv2-S/14	-0.15	0.00 (4)	-0.18	0.21 (5)	0.49 (4)	0.95	0.71 (5)	0.51 (5)
DINOv2-B/14	-0.45	0.00 (3)	0.18	0.46 (6)	0.51 (6)	0.92	0.76 (6)	0.59 (5)
CLIP ViT-B/16	<-1	0.00 (2)	0.33	0.58 (6)	0.68 (8)	0.25	0.83 (8)	0.64 (9)
SigLIP-B/16	-0.45	-0.00 (0)	0.05	0.48 (3)	0.60 (5)	0.56	0.77 (6)	0.59 (3)
MAE ViT-B/16	<-1	0.00 (4)	0.47	0.47 (12)	0.45 (9)	0.97	0.74 (8)	0.55 (12)
9 geometry features		0.78			0.44	0.97	0.813	0.78

fold by person-grouped CV. Under this procedure, CLIP selected layer 8 in four folds and layer 6 in one, and reached AUROC 0.827 (fold SD 0.016). DINOv2-B reached 0.777, SigLIP 0.769, DINOv2-S 0.732 and MAE 0.727. Geometry, run through the identical procedure, scored 0.816 (0.014). CLIP's paired difference from geometry was +0.011 (fold SD 0.019; positive in four of five folds), which five folds cannot distinguish from zero. The other models trailed geometry by 0.04–0.09, and DINOv2-S, SigLIP and MAE did so in every fold. Two caveats apply. The eye boxes come from the same MediaPipe landmarks that the geometry features use, and the readout is trained on continuous gaze angles. So this is evidence that the information is present, not a label-free alternative to geometry. Gaze is linearly present in mid-level patch tokens of every model we probed. A generic classifier on the summary token, the common way of using these models, cannot reach it.

6 Study 3: asking a generative vision-language model

Design (pre-registered). CLIP scores images against fixed captions; a generative VLM can instead be asked the question in plain language. We asked Qwen2.5-VL-3B-Instruct, the smallest current open model that fits a 4 GB laptop GPU (loaded in 4-bit), "Is this person looking directly at the camera? Answer with Yes or No." The input was the same face crop as every Study 1 method. The score was the logit of "Yes" minus that of "No" at the first answer token. Zero-shot (S3-a) covered all 5,878 images under the Study 1 folds and bootstrap. In the few-shot condition (S3-b), two labelled examples per class from the fold's training people were placed earlier in the conversation. These were the first two of Study 1's k=5 sample. S3-b used every positive and four negatives per positive in each test fold (1,400 images). Prompt, example order and subset were fixed before any output was seen.

Zero-shot, the model is the best label-free method, but only just. Asked directly, the VLM reached AUROC 0.601 [0.568, 0.633] (Fig. 12, left). This is better than CLIP's two prompts (0.547; paired difference +0.054 [0.023, 0.080]) and level with the L2CS-Net angle score (0.595; +0.006 [−0.033, 0.040]). It is also above geometry trained on five labels per class (0.537). Taken as a yes/no detector it is poor: it said "Yes" to 27% of all images, where only 4.8% are positive, for a balanced accuracy of 0.568.

What the model answers. The "Yes" rate follows head pose more than gaze (Fig. 12, right). For people looking at the camera, it fell from 71% with a frontal head to 16–18% at ±30°; for all other images, from 48% to 11–15%. On the 1,400-image subset the model said "Yes" to 40% of camera-directed images and to 38% of those 5° off (centre). Like every other method here, it cannot resolve the hardest distinction. The model appears to answer partly "is the face turned towards me?", which is the same head-pose shortcut

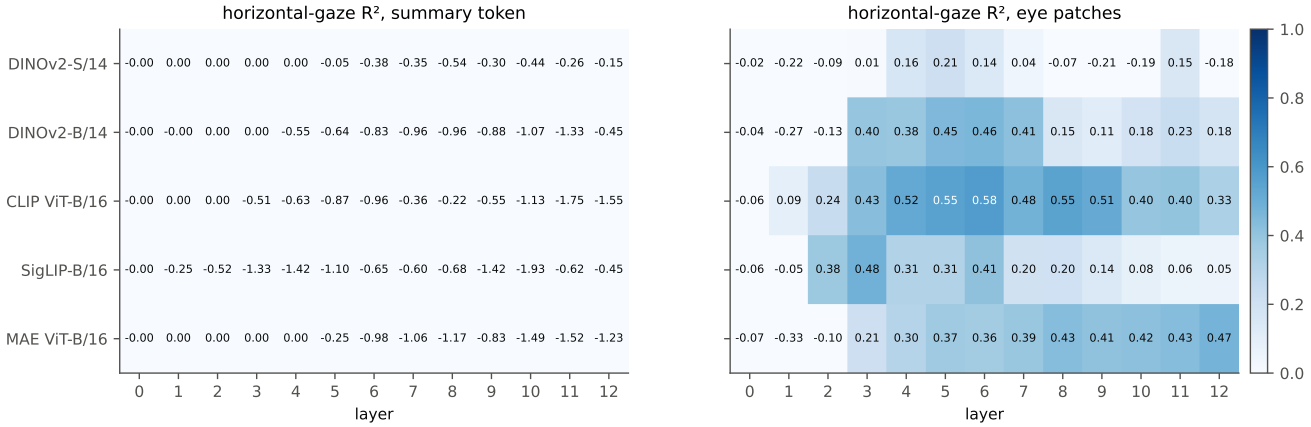


Figure 10: Horizontal-gaze R^2 for every model and layer. Left: summary token (never above 0.00). Right: eye-patch tokens. Colour is clipped to $[0, 1]$.

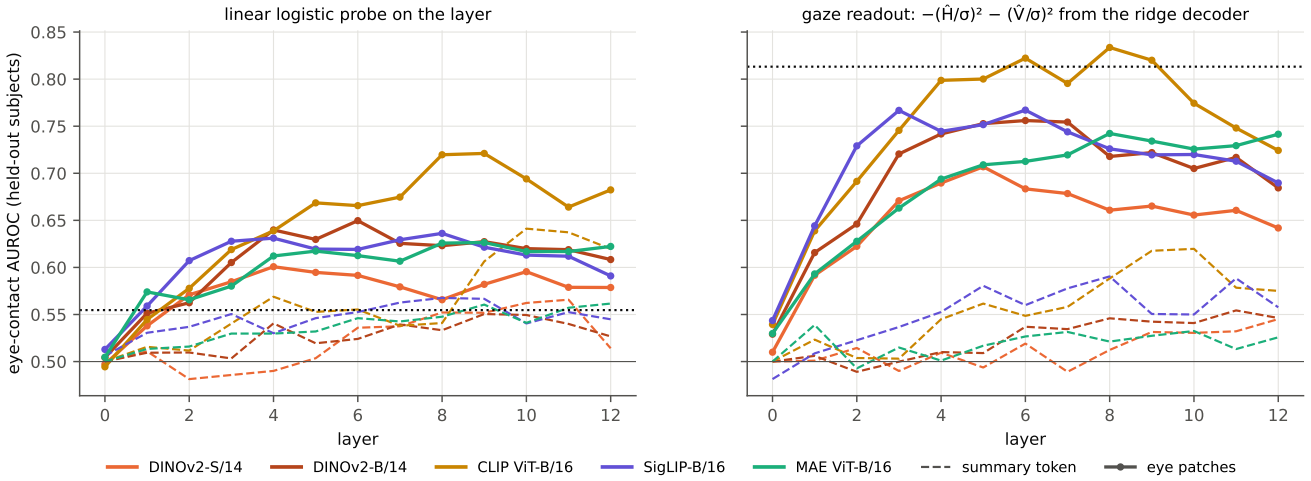


Figure 11: Eye-contact AUROC on held-out people by layer, mean over five folds. Left: linear logistic probe on binary labels. Right: gaze readout from the ridge decoder (trained with continuous gaze labels). Dotted lines show the geometry features under the same probe: 0.555 (left) and 0.813 (right).

that limits L2CS-Net.

In-context examples hurt. Two labelled examples per class lowered AUROC on the same images from 0.603 to 0.547 (paired difference −0.056 [−0.081, −0.026]). This is level with geometry trained on five labels per class (0.540 on this subset). The examples made the model say "Yes" less often (19%), but not more selectively: 21% for camera-directed images against 23% at 5°. Small VLMs are known to use in-context images weakly. Four examples of one fine-grained distinction do not seem to teach it; they mostly shift the model's bias. Larger models, other prompts and more examples could behave differently. We tested one of each, by design.

7 An in-browser demonstrator

We exported the geometry classifier trained on all 5,878 images and re-implemented the feature code in JavaScript on MediaPipe tasks-vision 0.10.21 (the same face-landmark model file as the Python pipeline). On 24 images resized exactly as in Python, the browser matched Python to 5.5×10^{-6} in landmark coordinates, 2×10^{-3} in features and 4.3×10^{-4} in probability, with no decision flips; the test also revealed that the library's transformation matrices are column-major, a detail whose mistreatment flipped 13 of 24 decisions. The demonstrator runs on camera frames or photos entirely on the client, flags inputs outside the training ranges, and states that it is a research artefact, not a detector.

8 Discussion

Under this protocol the most label-efficient representation was the simplest: nine geometric measurements with the right inductive bias for a target that sits at the centre of gaze space. Generic self-supervised embeddings were not label-efficient here because their dominant variation is identity and head pose, both irrelevant to the task. The gaze estimator shows the value and the limits of prior knowledge: it beats every method trained on ten labels or fewer without seeing a single FewGaze label, but its person- and pose-dependent bias caps what a pooled classifier can learn from its outputs. Study 2 locates the failure. Gaze is present in the mid-level eye-patch tokens of all five models, but the summary token drops it, and most models attenuate it in their final layers. The usual recipe,

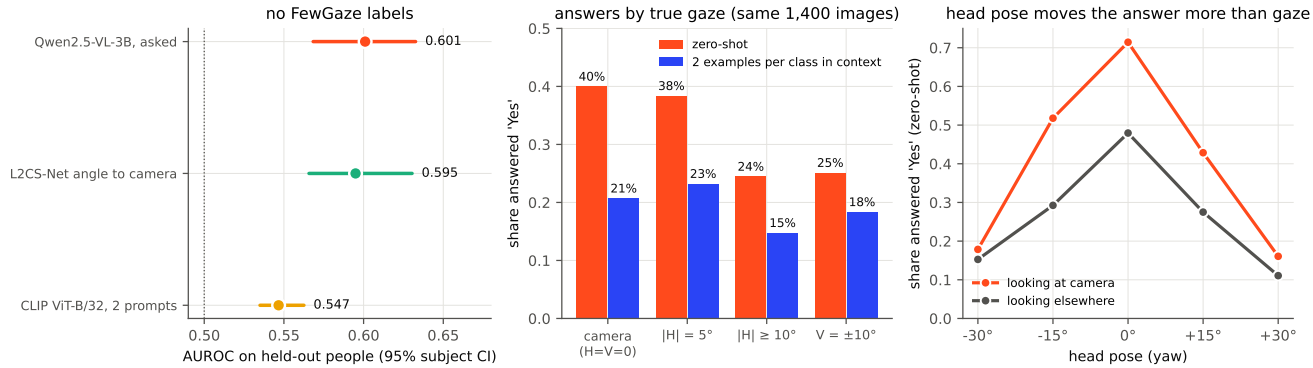


Figure 12: Study 3. Left: label-free methods on held-out people (fold-mean AUROC, 95% subject-bootstrap interval). Centre: share of "Yes" answers by true gaze, zero-shot and with two examples per class in context, on the same 1,400 images. Right: zero-shot "Yes" rate by head pose, for images looking at the camera and all others.

a linear probe on the last CLS token, therefore looks in the wrong place [5]. Study 3 adds that asking a small generative VLM is a reasonable zero-label baseline, and that a handful of in-context images does not turn it into a few-shot learner for this distinction. No method resolved the dataset's hardest distinction, 0° versus 5°, and absolute performance (AUROC ≤ 0.81) is far from deployable; it is not comparable to the per-image accuracy reported by Smith et al., who evaluated a different detector under a different protocol [21].

Towards interpersonal eye contact. This study is a deliberately controlled first step. The natural next step is social video with real eye-contact labels, such as the group conversations of Müller et al. [11, 12]. There, the target is another person, not the camera; heads move freely; and labels come per frame, not per posed photograph. We would carry over three things unchanged: subject-disjoint folds with nested label budgets, a subject-cluster bootstrap, and a pre-registered analysis plan. Studies 1 and 2 make two concrete predictions for that setting. First, eye-local representations (the geometry features, or mid-layer eye-patch tokens) should beat global embeddings. Second, a small amount of per-person calibration should matter more than more pooled labels, because the remaining error is person-specific. Testing both needs temporal models and target-relative gaze, which this dataset cannot provide.

9 Limitations and ethics

Results hold for one controlled dataset: 56 people, one lab, 2 m, yaw-only head pose, a chin rest. Camera-directed gaze is a proxy, not interpersonal eye contact, and individual fixation accuracy is unknown. Intervals capture subject sampling with models held fixed. Heads were fixed in advance; other probes might rank the embeddings differently. Study 3 tested one small VLM (3B parameters, 4-bit), one prompt and one set of examples; larger or full-precision models may do better. The dataset has no demographic labels, so we make no fairness claims, and pretrained models carry the biases of their training data. The dataset licence forbids redistribution, so no participant image appears in this paper. Gaze sensing can be misused for surveillance or attention scoring; our models are research artefacts and are unsuitable for monitoring or high-stakes use.

10 Conclusion

A handful of interpretable measurements with a quadratic boundary made the best use of labels, from 5 per class to several thousand; a frozen generic embedding with a linear probe did not learn the task, and a pretrained gaze estimator mainly helped when no labels were available. Inside foundation models, gaze is not absent but misplaced. Mid-layer eye-patch tokens carry it, and read out through decoded gaze, CLIP's tokens match nine hand-built measurements. Asked in words, a small generative VLM is the best label-free method, but it answers mostly about the head, and its examples do not help. The open question is whether person-specific calibration or eye-local representations can resolve the last 5° that separate looking *at* the camera from looking just past it.

Reproducibility. Every number, table and figure is regenerated by the commands in the repository from the recorded runs; the plans for all three studies were committed before they ran; images are not redistributed.

References

[1] A. A. Abdelrahman, T. Hempel, A. Khalifa, A. Al-Hamadi, and L. Dinges. L2CS-Net: Fine-grained gaze estimation in unconstrained environments. In *2023 8th International Conference on Frontiers of Signal Processing (ICFSP)*, pages 98–102, Corfu, Greece, 2023. IEEE. doi: 10.1109/ICFSP59764.2023.10372944. Preprint: arXiv:2203.03339.

[2] A. Ablavatski, A. Vakunov, I. Grishchenko, K. Raveendran, and M. Zhdanovich. Real-time pupil tracking from monocular video for digital puppetry, 2020. URL `https://arxiv.org/abs/2006.11341`.

[3] M. Bâce, S. Staal, and A. Bulling. Quantification of users' visual attention during everyday mobile device interactions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI)*, pages 1–14. ACM, 2020. doi: 10.1145/3313831.3376449.

[4] Y. Cheng, H. Wang, Y. Bao, and F. Lu. Appearance-based gaze estimation with deep learning: A review and benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46 (12):7509–7528, 2024. doi: 10.1109/TPAMI.2024.3393571. arXiv:2104.12668.

[5] M. El Banani, A. Raj, K.-K. Maninis, A. Kar, Y. Li, M. Rubinstein, D. Sun, L. Guibas, J. Johnson, and V. Jampani. Probing

the 3D awareness of visual foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 21795–21806, 2024. doi: 10.1109/CVPR52733.2024.02059.

[6] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15979–15988, 2022. doi: 10.1109/CVPR52688.2022.01553.

[7] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann. Real-time facial surface geometry from monocular video on mobile GPUs, 2019. URL https://arxiv.org/abs/1907.06724. CVPR Workshop on Computer Vision for Augmented and Virtual Reality.

[8] P. Kellnhofer, A. Recasens, S. Stent, W. Matusik, and A. Torralba. Gaze360: Physically unconstrained gaze estimation in the wild. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6911–6920, 2019. doi: 10.1109/ICCV.2019.00701.

[9] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann. MediaPipe: A framework for building perception pipelines, 2019. URL https://arxiv.org/abs/1906.08172.

[10] P. Müller, M. X. Huang, and A. Bulling. Detecting low rapport during natural interactions in small groups from non-verbal behaviour. In *Proceedings of the 23rd International Conference on Intelligent User Interfaces (IUI)*, pages 153–164, Tokyo, Japan, 2018. ACM. doi: 10.1145/3172944.3172969.

[11] P. Müller, M. X. Huang, X. Zhang, and A. Bulling. Robust eye contact detection in natural multi-person interactions using gaze and speaking behaviour. In *Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications (ETRA)*, pages 1–10, Warsaw, Poland, 2018. ACM. doi: 10.1145/3204493.3204549.

[12] P. Müller, M. Dietz, D. Schiller, D. Thomas, G. Zhang, P. Gebhard, E. André, and A. Bulling. MultiMediate: Multimodal group behaviour analysis for artificial mediation. In *Proceedings of the 29th ACM International Conference on Multimedia (MM)*, pages 4878–4882. ACM, 2021. doi: 10.1145/3474085.3479219.

[13] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jegou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=a68SUt6zFt. Featured Certification. arXiv:2304.07193.

[14] S. Park, X. Zhang, A. Bulling, and O. Hilliges. Learning to find eye region landmarks for remote gaze estimation in unconstrained settings. In *Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications (ETRA)*, pages 1–10, Warsaw, Poland, 2018. ACM. doi: 10.1145/3204493.3204545.

[15] S. Park, S. De Mello, P. Molchanov, U. Iqbal, O. Hilliges, and J. Kautz. Few-shot adaptive gaze estimation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9367–9376, Seoul, Korea, 2019. doi: 10.1109/ICCV.2019.00946.

[16] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011. URL https://jmlr.org/papers/v12/pedregosa11a.html.

[17] J. Qin, X. Zhang, and Y. Sugano. UniGaze: Towards universal gaze estimation via large-scale pre-training. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 5809–5820, 2026. doi: 10.1109/WACV61042.2026.00563. arXiv:2502.02307.

[18] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In M. Meila and T. Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html.

[19] F. Ryan, A. Bati, S. Lee, D. Bolya, J. Hoffman, and J. M. Rehg. Gaze-LLE: Gaze target estimation via large-scale learned encoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28874–28884, 2025. doi: 10.1109/CVPR52734.2025.02689. arXiv:2412.09586.

[20] Y. P. Shkolnikov. Do foundation models know geometry? probing frozen features for continuous physical measurement, 2026. URL https://arxiv.org/abs/2603.06459.

[21] B. A. Smith, Q. Yin, S. K. Feiner, and S. K. Nayar. Gaze locking: Passive eye contact detection for human-object interaction. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 271–280, St Andrews, Scotland, United Kingdom, 2013. ACM. doi: 10.1145/2501988.2501994.

[22] Y. Sugano, Y. Matsushita, and Y. Sato. Learning-by-synthesis for appearance-based 3D gaze estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1821–1828, Columbus, OH, USA, 2014. doi: 10.1109/CVPR.2014.235.

[23] Y. Sugano, X. Zhang, and A. Bulling. AggreGaze: Collective estimation of audience attention on public displays. In *Proceedings of the 29th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 821–831, Tokyo, Japan, 2016. ACM. doi: 10.1145/2984511.2984536.

[24] P. Yin, G. Zeng, J. Wang, and D. Xie. CLIP-Gaze: Towards general gaze estimation via visual-linguistic model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 6729–6737, 2024. doi: 10.1609/aaai.v38i7.28496.

[25] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11941–11952, 2023. doi: 10.1109/ICCV51070.2023.01100.

[26] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling. Appearance-based gaze estimation in the wild. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4511–4520, 2015. doi: 10.1109/CVPR.2015.7299081.

[27] X. Zhang, Y. Sugano, and A. Bulling. Everyday eye contact detection using unsupervised gaze target discovery. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 193–203, Québec City, QC, Canada, 2017. ACM. doi: 10.1145/3126594.3126614.

[28] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling. It's written all over your face: Full-face appearance-based gaze estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 2299–2308, Honolulu, HI, USA, 2017. doi: 10.1109/CVPRW.2017.284.

[29] X. Zhang, Y. Sugano, and A. Bulling. Revisiting data normal-

ization for appearance-based gaze estimation. In *Proceedings of the 2018 ACM Symposium on Eye Tracking Research & Applications (ETRA)*, pages 1–9, Warsaw, Poland, 2018. ACM. doi: 10.1145/3204493.3204548.

[30] X. Zhang, Y. Sugano, and A. Bulling. Evaluation of appearance-based methods and implications for gaze-based applications. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI)*, pages 1–13, Glasgow, Scotland, UK, 2019. ACM. doi: 10.1145/3290605.3300646.

[31] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling. MPIIGaze: Real-world dataset and deep appearance-based gaze estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(1):162–175, 2019. doi: 10.1109/TPAMI.2017.2778103.

[32] X. Zhang, S. Park, T. Beeler, D. Bradley, S. Tang, and O. Hilliges. ETH-XGaze: A large scale dataset for gaze estimation under extreme head pose and gaze variation. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, pages 365–381. Springer International Publishing, 2020. doi: 10.1007/978-3-030-58558-7_22.

A Supplementary figures

These figures support Section 4 and the demonstrator section. All of them come from the same verified runs as the main text.

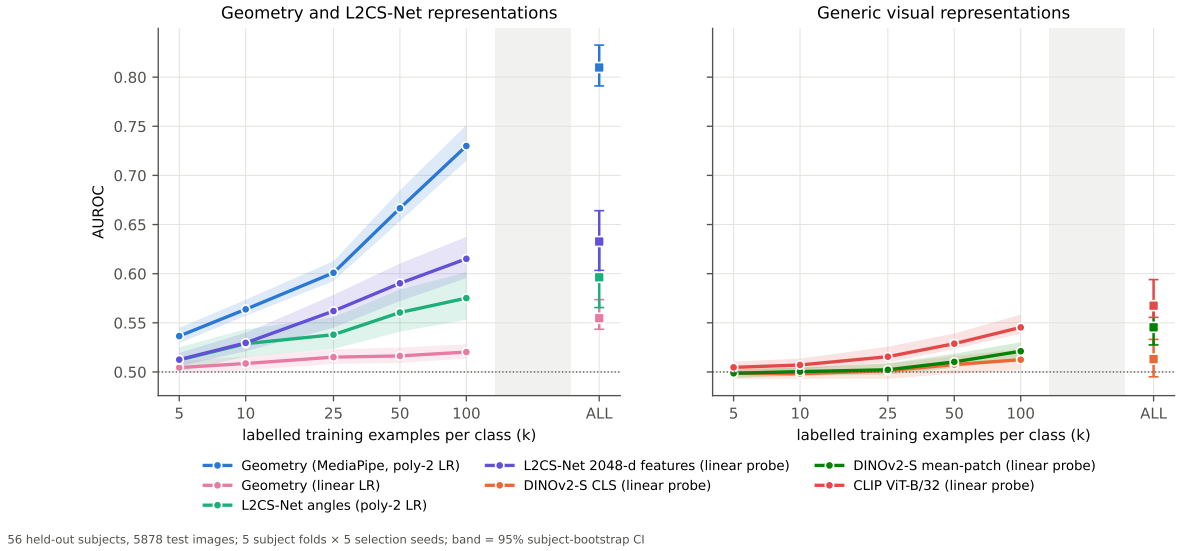


Figure 13: Ablations: representation choices within each family.

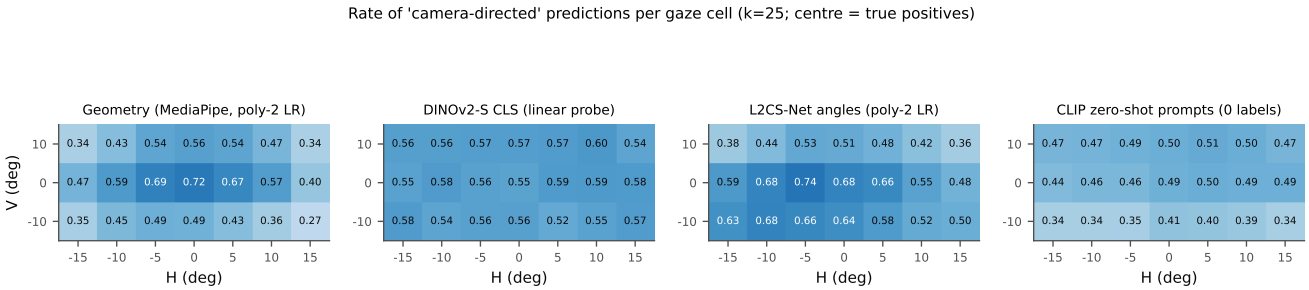


Figure 14: Share of images predicted camera-directed per gaze cell at k=25 (centre = true positives).

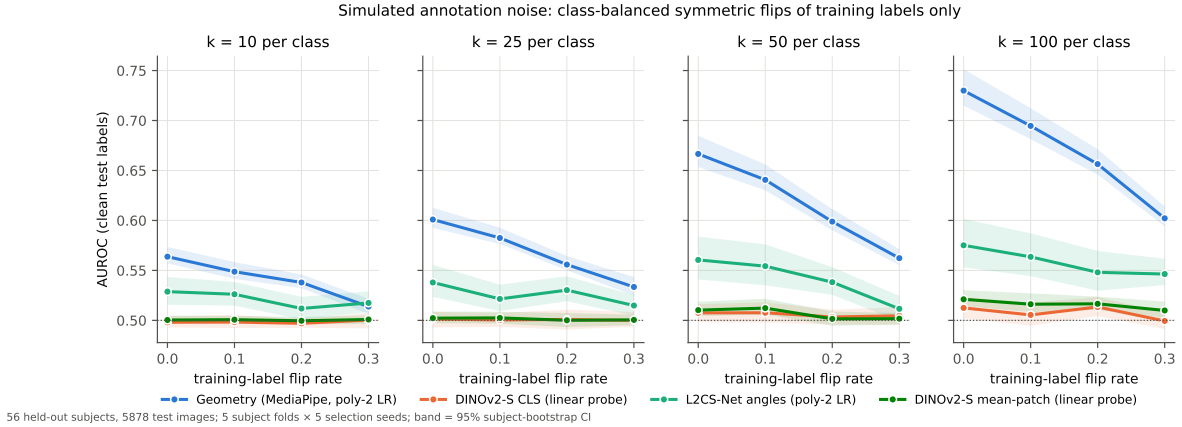


Figure 15: Simulated label noise: class-balanced flips of training labels only, identical across methods.

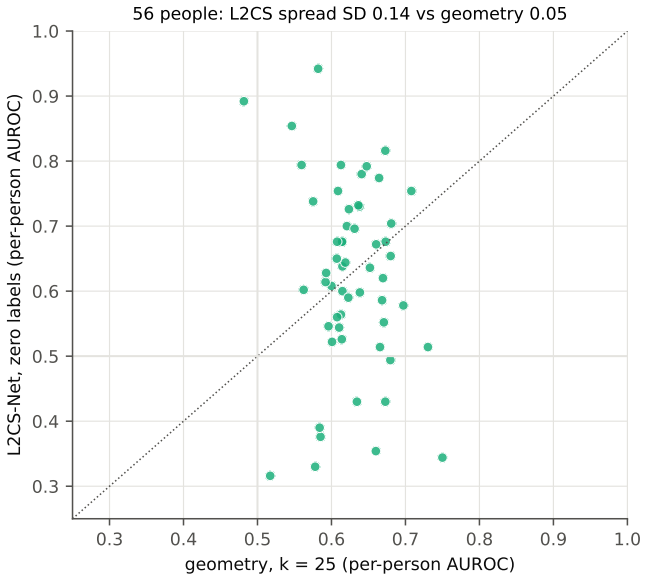


Figure 16: Per-person AUROC: L2CS-Net without labels versus geometry at k=25.

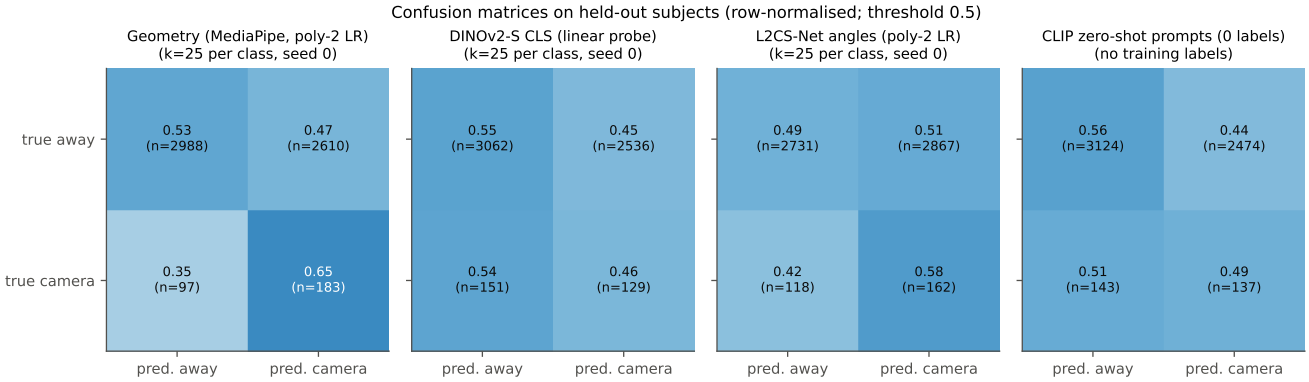


Figure 17: Confusion matrices on held-out people (row-normalised), k=25, seed 0; CLIP uses no labels.

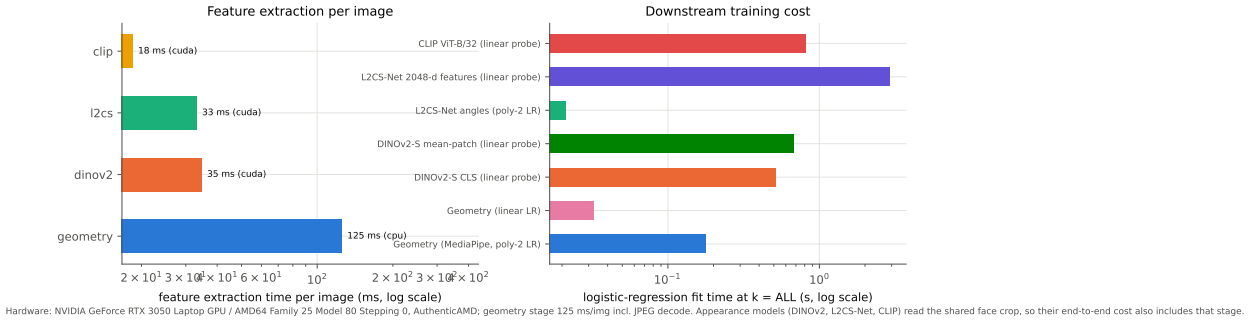


Figure 18: Feature-extraction and training cost on one laptop (RTX 3050 laptop GPU).